Himaayah Schools
ACADEMY

FREE GUIDE

وَالسَّمَاءَ رَفَعَهَا وَوَضَعَ الْمِيزَانَ

"And the heaven He raised, and He set up the balance." (ar-Raḥmān 55:7)

JAMB Physics

A UTME study guide: the syllabus, the formulas you must know, worked examples, common traps and a practice test.

بِسْمِ اللَّهِ الرَّحْمٰنِ الرَّحِيمِ

Before you begin

Physics in the UTME rewards two things: understanding the ideas, and practising calculations until they are fast. Most questions are short calculations or questions on definitions and laws. This guide gives you the syllabus in six areas, a formula sheet with units, nine worked examples, the traps our mentors see most often, an 8-week plan and a practice test.

The UTME at a glance	
Subjects	Use of English and three subjects for your course. Physics is required for many science, medical and engineering courses.
Questions	Use of English 60; each other subject 40; 180 questions in all, on computer (CBT).
Time	About two hours for all four papers, so roughly 40 seconds a question. You move freely between subjects.
Marking	Each subject is scored out of 100, for a total out of 400. There is no negative marking: never leave a question blank.

Check the official details

Exam rules and syllabuses can change from year to year. Always check the current JAMB brochure and the official Physics syllabus on the JAMB website or e-facility portal, and follow them where they differ from this guide.

Contents

01 **The syllabus at a glance**
What JAMB can ask, in six areas

02 **The formula sheet**
Equations and units you must know by heart

03 **Worked examples**
Step by step, the way the exam asks

04 **Traps that cost marks**
And how to avoid them

05 **An 8-week plan**
From the syllabus to the exam hall

06 **Practice test and answers**
15 questions

07 **Recommended textbooks**

Part 1 The syllabus at a glance

Study from the **JAMB syllabus**, not only from your school notes: every UTME question comes from a topic on it. The topics fall into these areas:

Area	Topics
Measurement and mechanics	Units and dimensions; scalars and vectors; motion (linear, projectile, circular, simple harmonic); Newton's laws; momentum; gravitational field; equilibrium of forces and moments; work, energy and power; friction; simple machines; elasticity
Matter and fluids	Pressure in solids, liquids and gases; Archimedes' principle and flotation; density and relative density; structure of matter and kinetic theory; surface tension and capillarity
Heat	Temperature and thermometers; thermal expansion; gas laws; specific heat capacity and latent heat; change of state; vapours; heat transfer
Waves and optics	Wave properties; sound and its characteristics; resonance; light: reflection at plane and curved mirrors, refraction at plane surfaces and lenses; total internal reflection; optical instruments; dispersion and colour; the electromagnetic spectrum
Electricity and magnetism	Electrostatics; capacitors; cells; current electricity and Ohm's law; electrical energy and power; magnets and fields; force on a current-carrying conductor; electromagnetic induction; transformers; simple a.c. circuits; conduction through liquids and gases
Modern physics and electronics	Structure of the atom; photoelectric effect; X-rays; radioactivity and half-life; nuclear reactions; wave-particle duality; semiconductors, diodes and simple electronics

Mechanics and electricity together usually make up the largest share of the paper, so give them the most time, but do not skip any area: every area appears every year.

Part 2 The formula sheet

Mechanics

Quantity	Formula	Unit
Equations of motion	$v = u + at$; $s = ut + \frac{1}{2}at^2$; $v^2 = u^2 + 2as$; $s = \frac{1}{2}(u + v)t$	m, m/s, m/s²
Free fall (up/down)	Max height $H = u^2/2g$; time to top $t = u/g$	$g \approx 10$ m/s²
Projectile	Range $R = u^2 \sin 2\theta / g$; time of flight $T = 2u \sin\theta / g$; $H = u^2 \sin^2\theta / 2g$	m, s
Newton's 2nd law, weight	$F = ma$; $W = mg$	N
Momentum, impulse	$p = mv$; impulse $= Ft = mv - mu$	kg m/s = N s
Work, energy, power	$W = Fs$; $KE = \frac{1}{2}mv^2$; $PE = mgh$; $P = W/t = Fv$	J, W
Circular motion	$a = v^2/r$; $F = mv^2/r$; $v = \omega r$	m/s², N
Pendulum, spring	$T = 2\pi\sqrt{(l/g)}$; $T = 2\pi\sqrt{(m/k)}$	s
Gravitation	$F = Gm_1m_2/r^2$; $g = GM/R^2$	N
Machines	MA = load/effort; VR = distance moved by effort/distance moved by load; efficiency = MA/VR × 100%	—
Hooke's law, Young's modulus	$F = ke$; E = stress/strain $= (F/A)/(e/l)$	N/m, N/m²
Pressure	$P = F/A$; in a liquid $P = h\rho g$	Pa = N/m²

Heat

Quantity	Formula	Unit
Heat capacity, latent heat	$Q = mc\theta$; $Q = mL$	J
Linear expansivity	$\alpha = (L_2 - L_1)/(L_1\theta)$; area $\beta = 2\alpha$; volume $\gamma = 3\alpha$	K^{-1}
Gas laws	Boyle PV = constant; Charles V/T = constant; pressure law P/T = constant; general $P_1V_1/T_1 = P_2V_2/T_2$ (T in kelvin)	$K = °C + 273$

Waves and optics

Quantity	Formula	Unit
Wave equation	$v = f\lambda$; $T = 1/f$	m/s, Hz
Mirrors and lenses	$1/f = 1/u + 1/v$; $m = v/u$ = image height/object height; power = 1/f (f in metres)	m, dioptre
Refraction	$n = \sin i / \sin r$ = real depth / apparent depth; $\sin C = 1/n$	—
Pipes and strings	Closed pipe: $L = \lambda/4$ (fundamental); open pipe and string: $L = \lambda/2$; string $f = (1/2L)\sqrt{(T/\mu)}$	m, Hz

Electricity and modern physics

Quantity	Formula	Unit
Coulomb's law, field	$F = kq_1q_2/r^2$; $E = F/q = V/d$	N, N/C
Capacitors	$C = Q/V$; $C = \varepsilon A/d$; energy = $\frac{1}{2}CV^2$; parallel $C = C_1 + C_2$; series $1/C = 1/C_1 + 1/C_2$	F, J
Ohm's law, resistors	$V = IR$; series $R = R_1 + R_2$; parallel $1/R = 1/R_1 + 1/R_2$; $R = \rho l/A$	Ω
E.m.f.	$E = I(R + r)$; terminal p.d. $V = IR$	V
Power, energy	$P = IV = I^2R = V^2/R$; energy = Pt; 1 kWh = 3.6×10^6 J	W, J
Electrolysis	$m = zIt$ (Faraday's first law)	kg
Transformer	$V_s/V_p = N_s/N_p = I_p/I_s$ (ideal)	—
A.c. circuits	$V_{rms} = V_0/\sqrt{2}$; $X_L = 2\pi fL$; $X_C = 1/(2\pi fC)$; $Z = \sqrt{(R^2 + (X_L - X_C)^2)}$; resonance $f_0 = 1/(2\pi\sqrt{(LC)})$	Ω, Hz
Photons, photoelectric effect	$E = hf = hc/\lambda$; $hf = W_0 + \frac{1}{2}mv^2_{max}$; $\lambda = h/p$	J, eV
Half-life	$N = N_0(\frac{1}{2})^{(t/T\frac{1}{2})}$	—

Learn the units with the formulas: a quick unit check catches many errors. Prefixes: milli (m) = 10^{-3}, micro (μ) = 10^{-6}, nano (n) = 10^{-9}, kilo (k) = 10^3, mega (M) = 10^6. 1 eV = 1.6×10^{-19} J.

Part 3 Worked examples

Example 1. A car accelerates uniformly from 10 m/s to 30 m/s in 5 s. Find its acceleration and the distance covered.

→ $a = (v - u)/t = (30 - 10)/5 = 4$ m/s²

→ $s = \frac{1}{2}(u + v)t = \frac{1}{2}(10 + 30) \times 5 = 100$ m

Answer: 4 m/s² and 100 m

Example 2. A ball is thrown vertically upward at 20 m/s. Find the maximum height and the total time in the air. ($g = 10\ m/s^2$)

→ $H = u^2/2g = 400/20 = 20$ m

→ Time to the top = $u/g = 2$ s; total time = $2 \times 2 = 4$ s

Answer: 20 m; 4 s

Example 3. Resistors of 2 Ω, 3 Ω and 6 Ω are connected in parallel across a cell of e.m.f. 6 V and internal resistance 0.5 Ω. Find the current and the terminal p.d.

→ $1/R = 1/2 + 1/3 + 1/6 = 1$, so $R = 1\ \Omega$

→ $I = E/(R + r) = 6/1.5 = 4$ A

→ $V = IR = 4 \times 1 = 4$ V

Answer: 4 A; 4 V

Example 4. An object is placed 15 cm from a concave mirror of focal length 10 cm. Find the image position and magnification.

→ $1/v = 1/f - 1/u = 1/10 - 1/15 = 1/30$

→ $v = 30$ cm (real, in front of the mirror)

→ $m = v/u = 30/15 = 2$

Answer: Image 30 cm from the mirror, real, inverted, twice the size

Example 5. How much heat is needed to raise 0.5 kg of water from 20 °C to 100 °C? ($c = 4200$ J/kg K)

→ $Q = mc\theta = 0.5 \times 4200 \times 80$

→ $Q = 168\,000$ J

Answer: 168 kJ

Example 6. A radioactive sample of 80 g has a half-life of 5 days. How much remains after 15 days?

→ 15 days = 3 half-lives

→ 80 → 40 → 20 → 10 g

Answer: 10 g

Example 7. A transformer has 200 turns on the primary and 1000 on the secondary. The primary is connected to 240 V a.c. Find the secondary voltage.

→ $Vs/Vp = Ns/Np$, so $Vs = 240 \times 1000/200$

→ $Vs = 1200$ V (a step-up transformer)

Answer: 1200 V

Example 8. A sound wave has a frequency of 500 Hz and a wavelength of 0.68 m. Find its speed.

→ $v = f\lambda = 500 \times 0.68$

Answer: 340 m/s

Example 9. A gas occupies 100 cm^3 at 1 atm. At constant temperature, it is compressed to 2 atm. Find the new volume.

→ Boyle's law: $P_1V_1 = P_2V_2$

→ $V_2 = (1 \times 100)/2 = 50\ cm^3$

Answer: 50 cm^3

Part 4 Traps that cost marks

Trap	What to do
Mixing units (cm with m, g with kg, minutes with seconds)	Convert everything to SI units before you substitute. Write the unit with every answer.
Using °C in the gas laws	Always convert to kelvin: T = θ + 273.
Forgetting internal resistance	If a cell has internal resistance r, the current is E/(R + r), and the terminal p.d. is less than the e.m.f.
Sign errors with mirrors and lenses	Use one sign convention consistently (real is positive is common in Nigerian textbooks). A virtual image gives a negative v.
Confusing mass and weight	Mass is in kg and does not change; weight W = mg is a force in newtons.
Half-life arithmetic	Count the number of half-lives first (t ÷ T½), then halve that many times.
kWh and joules	1 kWh = 1000 W × 3600 s = 3.6×10^6 J. Cost = energy in kWh × price per kWh.
Vectors added as scalars	Forces and velocities at an angle are added with Pythagoras or resolution, not simple addition.
Not reading "which is NOT"	Underline NOT, EXCEPT and LEAST in the question before you choose.

Part 5 An 8-week plan

Start at least eight weeks before the exam. Each week: learn the topics, make short notes and solve past questions under time. Keep one day a week for revision.

Week	Focus	Practice
1	Units, vectors, motion, Newton's laws, momentum	Equations of motion: 30 questions
2	Work, energy, power, machines, friction, elasticity, pressure, fluids	Machines and pressure problems
3	Gravitation, circular motion, SHM, equilibrium and moments	Mixed mechanics past questions
4	Heat: expansion, gas laws, heat capacity, latent heat, transfer	Heat calculations; timed set
5	Waves, sound, light: mirrors, refraction, lenses, instruments	Mirror and lens formula drills
6	Electrostatics, capacitors, current electricity, power	Circuits: series, parallel, e.m.f.
7	Magnetism, induction, transformers, a.c., modern physics, electronics	Half-life and photoelectric questions
8	Revision of weak areas; full timed mock exams	Two full CBT mocks; review every error

Exam-day reminders

Read every option before choosing. Eliminate the clearly wrong ones first.

Do not spend more than a minute on one question: flag it and come back.

Answer every question; there is no penalty for a wrong answer.

Check that your answers are saved before you submit, and keep an eye on the clock for all four subjects.

Test Practice test

Give yourself 10 minutes for these 15 questions, as in the real exam. The answers follow.

1. Which of these is a vector quantity?

a) Speed b) Mass c) Momentum d) Energy

2. A body starts from rest and accelerates at 2 m/s² for 5 s. Its final velocity is:

a) 2.5 m/s b) 10 m/s c) 7 m/s d) 25 m/s

3. The SI unit of power is the:

a) Joule b) Newton c) Pascal d) Watt

4. A machine has a velocity ratio of 5 and an efficiency of 80%. Its mechanical advantage is:

a) 4 b) 6.25 c) 0.16 d) 400

5. The pressure at a depth of 10 m in water (ρ = 1000 kg/m³, g = 10 m/s²) due to the water is:

a) 10^3 Pa b) 10^4 Pa c) 10^5 Pa d) 10^6 Pa

6. Which mode of heat transfer does not need a material medium?

a) Conduction b) Convection c) Radiation d) Evaporation

7. The temperature 27 °C is equal to:

a) 246 K b) 273 K c) 300 K d) 327 K

8. Sound cannot travel through:

a) A vacuum b) Water c) Steel d) Air

9. The critical angle of a glass of refractive index 2 is:

a) 60° b) 45° c) 30° d) 90°

10. Two 4 Ω resistors in parallel have an effective resistance of:

a) 8 Ω b) 4 Ω c) 1 Ω d) 2 Ω

11. A 100 W bulb used for 10 hours consumes:

a) 1 kWh b) 10 kWh c) 0.1 kWh d) 1000 kWh

12. A step-down transformer:

a) Increases current and voltage b) Has more turns on the secondary c) Works on d.c. d) Has fewer turns on the secondary

13. After 2 half-lives, the fraction of a radioactive sample remaining is:

a) 1/2 b) 1/8 c) 1/4 d) 1/16

14. The photoelectric effect shows that light behaves as:

a) Waves only b) Particles (photons) c) Sound d) A longitudinal wave

15. A p-n junction diode is mainly used for:

a) Rectification b) Amplifying sound c) Storing charge d) Measuring resistance

No.	Answer	No.	Answer
1	c) Momentum	9	c) 30°
2	b) 10 m/s	10	d) 2 Ω
3	d) Watt	11	a) 1 kWh
4	a) 4	12	d) Fewer secondary turns
5	c) 10^5 Pa	13	c) 1/4
6	c) Radiation	14	b) Particles
7	c) 300 K	15	a) Rectification
8	a) A vacuum		

Recommended textbooks

- **New School Physics for Senior Secondary Schools**, M. W. Anyakoha.
- **Senior Secondary Physics**, P. N. Okeke and M. W. Anyakoha.
- **Comprehensive Certificate Physics**, O. Awe and O. O. Ogunleye.
- Other texts listed in the current JAMB Physics syllabus.
- **Past questions**: solve at least the last ten years' UTME Physics questions, timing yourself, and study every one you get wrong.

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وَمَا تَوْفِيقِي إِلَّا بِاللَّهِ

"And my success is only through Allah." (Hūd 11:88)

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Understand it, then practise it.

This free guide from the Himaayah Academy helps UTME candidates prepare for Physics with a clear plan, a formula sheet and the mistakes our mentors see most.

- The JAMB Physics syllabus in six areas
- A formula sheet with units
- Nine worked examples, step by step
- The traps that cost marks
- An 8-week plan and a 15-question practice test

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